Capstone Project Report: E-Commerce Data Analysis

1. Introduction & Dataset Overview

- **Objective:** The goal of this capstone project is to clean, process, and analyze an e-commerce retail dataset to extract actionable business insights regarding sales performance, customer behavior, and product demand.
- **Dataset:** Online Retail dataset containing transaction records from a UK-based gift retailer.
- **Tools Used:** Python (Pandas, Matplotlib, Seaborn), VS Code, Jupyter Notebook.

```python
import pandas as pd

# Tumchya CSV file cha exact nav ithe dya
df = pd.read_csv(r"C:\Users\khond\Downloads\online-retail-dataset.csv")

# Data che pahile 5 rows baghnyasathi
print(df.head())
```

```
...    InvoiceNo StockCode                          Description  Quantity  \
    0    536365    85123A   WHITE HANGING HEART T-LIGHT HOLDER         6
    1    536365     71053                  WHITE METAL LANTERN         6
    2    536365    84406B       CREAM CUPID HEARTS COAT HANGER         8
    3    536365    84029G  KNITTED UNION FLAG HOT WATER BOTTLE         6
    4    536365    84029E       RED WOOLLY HOTTIE WHITE HEART.         6

          InvoiceDate  UnitPrice  CustomerID         Country
    0  12/1/2010 8:26       2.55     17850.0  United Kingdom
    1  12/1/2010 8:26       3.39     17850.0  United Kingdom
    2  12/1/2010 8:26       2.75     17850.0  United Kingdom
    3  12/1/2010 8:26       3.39     17850.0  United Kingdom
    4  12/1/2010 8:26       3.39     17850.0  United Kingdom
    <class 'pandas.DataFrame'>
    RangeIndex: 541909 entries, 0 to 541908
    Data columns (total 8 columns):
     #   Column       Non-Null Count   Dtype
    ---  ------       --------------   -----
     0   InvoiceNo    541909 non-null  str
     1   StockCode    541909 non-null  str
     2   Description  540455 non-null  str
     3   Quantity     541909 non-null  int64
     4   InvoiceDate  541909 non-null  str
     5   UnitPrice    541909 non-null  float64
     6   CustomerID   406829 non-null  float64
     7   Country      541909 non-null  str
    dtypes: float64(2), int64(1), str(5)
    memory usage: 33.1 MB
    None
```

2. Data Cleaning & Preprocessing

Raw data often contains anomalies, missing values, and duplicates that can distort analysis. The following cleaning steps were executed:

- **Missing Value Treatment:** Checked null values across all columns. Rows with missing CustomerID were dropped, and missing descriptions were filled with 'Unknown'.
- **Noise & Cancellation Removal:** Filtered out negative or zero values in Quantity and UnitPrice corresponding to product returns and cancellations.
- **Dataset Reduction:** Cleaned the dataset from initial 5,41,909 records down to 3,92,692 valid transaction records.

```
##Step 2 Data Cleaning



# 1. Kontya column madhe kiti missing values ahet te check karne
print("Missing values in each column:")
print(df.isnull().sum())

# 2. Duplicate rows remove karne
df = df.drop_duplicates()

# 3. CustomerID madhlya missing values la drop kinva fill karu shakta (sadhya drop karuya)
df = df.dropna(subset=['CustomerID'])
```

```
Missing values in each column:
InvoiceNo          0
StockCode          0
Description     1454
Quantity           0
InvoiceDate        0
UnitPrice          0
CustomerID    135080
Country            0
dtype: int64
```

```
##Step: Missing Values Handling and Data Cleaning



# 1. Missing CustomerID remove karne
df = df.dropna(subset=['CustomerID'])

# 2. Description madhlya missing values la 'Unknown' ne fill karne
df['Description'] = df['Description'].fillna('Unknown')

# 3. Negative quantity ya unit price asel (returns/cancellations) tar tyanna check karu shakta
# Sadhya fakt positive values filter karuya (jar sales analysis karaycha asel)
df = df[df['Quantity'] > 0]
df = df[df['UnitPrice'] > 0]

# 4. Cleaned rows kiti urlyat te check karne
print("Total rows after cleaning:", df.shape[0])
```

```
Total rows after cleaning: 392692
```

3. Feature Engineering

To enable advanced exploratory data analysis, new features were engineered from existing attributes:

- **TotalSales:** Calculated by multiplying Quantity and UnitPrice for each transaction line item.
- **Date Parsing:** Converted InvoiceDate from string format into standard Python datetime objects and extracted Year and Month components for temporal trend analysis.

```python
# 1. Total Sales column banavne (Quantity * UnitPrice)
df['TotalSales'] = df['Quantity'] * df['UnitPrice']

# 2. InvoiceDate la datetime format madhe convert karne
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])

# 3. Check karnyasathi pahile 5 rows print karne
print(df[['InvoiceNo', 'Quantity', 'UnitPrice', 'TotalSales', 'InvoiceDate']].head())
```

```
  InvoiceNo  Quantity  UnitPrice  TotalSales         InvoiceDate
0    536365         6       2.55       15.30 2010-12-01 08:26:00
1    536365         6       3.39       20.34 2010-12-01 08:26:00
2    536365         8       2.75       22.00 2010-12-01 08:26:00
3    536365         6       3.39       20.34 2010-12-01 08:26:00
4    536365         6       3.39       20.34 2010-12-01 08:26:00
```

4. Exploratory Data Analysis (EDA) & Visualizations

Data analysis was performed to uncover key business metrics and patterns:

- **Top Revenue Products:** Identified the top-performing products generating the highest gross revenue.
- **Geographic Insights:** Analyzed sales distribution across various countries, revealing core international markets.

```python
# Highest sales denare top 5 products
top_products = df.groupby('Description')['TotalSales'].sum().sort_values(ascending=False).head(5)
print("--- Top 5 Products by Revenue ---")
print(top_products)
```

```
--- Top 5 Products by Revenue ---
Description
PAPER CRAFT , LITTLE BIRDIE           168469.60
REGENCY CAKESTAND 3 TIER              142264.75
WHITE HANGING HEART T-LIGHT HOLDER    100392.10
JUMBO BAG RED RETROSPOT                85040.54
MEDIUM CERAMIC TOP STORAGE JAR         81416.73
Name: TotalSales, dtype: float64
```

##Step 3: Top 5 Countries (Revenue wise)

```python
# Kuthlya deshatun sarvat jast revenue ala
top_countries = df.groupby('Country')['TotalSales'].sum().sort_values(ascending=False).head(5)
print("\n--- Top 5 Countries by Revenue ---")
print(top_countries)
```

```
--- Top 5 Countries by Revenue --
Country
United Kingdom    7285024.644
Netherlands        285446.340
EIRE               265262.460
Germany            228678.400
France             208934.310
Name: TotalSales, dtype: float64
```

5. Key Findings & Conclusion

- **Revenue Drivers:** A small subset of top-tier products drives a significant portion of total retail sales.
- **Market Concentration:** The United Kingdom represents the largest market share for transaction volume and revenue.
- **Business Impact:** The cleaned dataset provides a robust foundation for inventory planning, targeted marketing, and sales forecasting.